Advanced Applied Statistics

Lecture 4: Bayesian Inference and Empirical Bayes

Prof. Sami S. Brandt

Department of Computer Science
IT University of Copenhagen

16 September 2025

IT UNIVERSITY OF COPENHAGEN

Learning Objectives of This Lecture

- Relate the use of Bayes rule within statistical inference
- Explain the frequentist and Bayesian interpretation of probability
- Relate the use of prior probability densities to characterise prior information
- Explain the concept and derive uninformative priors
- Compare frequentist and Bayesian inference
- Explain and apply the concept of empirical Bayes

Core of Bayesian Inference: the Bayes rule

Theorem

Let A and B be events, where $P(A) \neq 0$. Then

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}. \tag{1}$$

The equation yields a way to incorporate **prior information** into statistical models.

Physicist's Twins

Example

A physicist found out she was going to have twin boys.

Q (Phycisist): "What is the probability they are **Identical**, rather than **Fraternal**?"

A (Doctor): $1/3$ of twin births are identicals, $2/3$ fraternals.

Applying Bayes' rule yields

$$\frac{P(\text{Identical}|\text{Same sex})}{P(\text{Fraternal}|\text{Same sex})} = \frac{P(\text{Same sex} \mid \text{Identical})}{P(\text{Same sex} \mid \text{Fraternal})} \frac{P(\text{Identical})}{P(\text{Fraternal})} \tag{2}$$

$$= \frac{1}{1/2} \frac{1/3}{2/3} = 1. \tag{3}$$

$\Longrightarrow$ The odds of having fraternal or identical twins are even when one knows they have the same sex.

The twins problem

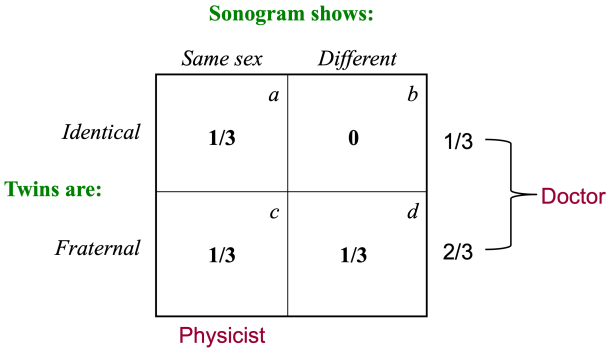


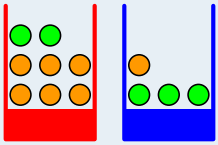
Figure 3.1 Analyzing the twins problem.

- Doctor's prior distribution most likely based on some enormous national database (frequentist approach)
- However, sometimes such previous experience is not available—there are *other* strategies to construction of an appropriate prior (Bayesian approach)

Bayesian Notion for Probability

Frequentist definition of probability

The probability of an event is **the number of times the event occurs out of the total number of trails**, as the number of trials goes to infinity:
e.g. $p(F = a) = \frac{n_a}{N}$ as $N \to \infty$



However, not all events can be repeated numerous times:



What is the probability of Denmark winning the World Cup in 2026?



What is the probability that the Arctic ice cap will have melted by the end of the century?

Probability as a measure of uncertainty

We can all agree that the probability of **heads** is $\frac{1}{2}$ for a toss of a fair coin.

Here are to common justifications

1. **Frequency argument:**

$$\text{probability} = \text{relative frequency obtained in a long sequence of tosses,}$$

   i.e. $P(H) = \frac{n_H}{N}$ as $N \to \infty$..

2. **Symmetry or exchangeability argument:**

$$\text{probability} = \frac{\text{number of favorable cases}}{\text{number of possibilities}}$$

   assuming equally likely possibilities, i.e. $P(H) = \frac{1}{2}$.

The frequents argument involves the hypothetical notion of a long sequence of experiments.

The symmetry argument allow us to assign probability to a singe experiment.



Picture from gurmeet.net

Bayesian Probabilities

Bayesian philosophy in a nutshell

- Probabilities do not have to be based on the frequencies of outcome of experiments (contrary to frequentists view-point)
- We may assign probabilities to e.g. parameters and models
- Probabilities express the degree of uncertainty about our knowledge of a variable

Bayesian inference

- Assume we want to learn parameters θ from the observations or data x.
- We may make inference about the parameters θ of a statistical model by utlising the **Bayes theorem**



$$\overbrace{p(\theta|x)}^{\text{posterior}} = \frac{\overbrace{p(x|\theta)}^{\text{likelihood}}\overbrace{p(\theta)}^{\text{prior}}}{\underbrace{p(x)}_{\text{evidence}}},$$



where $p(x) = \int_\Omega p(x|\theta)p(\theta)d\theta$, or

$$\text{posterior} \propto \text{likelihood} \times \text{prior}$$

Prior: Assumptions about θ *before* seeing the data.

Likelihood: The effect of the observed data x.

Posterior: The uncertainty in θ *after* observing x.

Pictures from wikipedia.org and mt-oceanography.info

The Bayesian method

1. Choose a prior density $p(\theta)$ on the parameters $\theta \in \Omega$.
2. Choose a statistical model $f_\theta(x) \equiv p(x|\theta)$ that reflects our beliefs about x given θ.
3. After observing data x, update our beliefs and calculate the posterior distribution $p(\theta|x)$.

Note: From Bayes' rule, $p(\theta|x) \propto L(\theta)p(\theta)$, where $L(\theta)$ is the likelihood function, that is, $f_\theta(x)$ with x fixed and θ varying.

Uncertainty of Correlation Coefficient

Example

Table 3.1 *Scores from two tests taken by 22 students,* **mechanics** *and* **vectors**.

	1	2	3	4	5	6	7	8	9	10	11
mechanics	7	44	49	59	34	46	0	32	49	52	44
vectors	51	69	41	70	42	40	40	45	57	64	61

	12	13	14	15	16	17	18	19	20	21	22
mechanics	36	42	5	22	18	41	48	31	42	46	63
vectors	59	60	30	58	51	63	38	42	69	49	63

The sample corrlelation coefficient between two scores is $\hat{\theta} = 0.498$
Assuming that the joint (m,v) distribution is bivariate normal

$$f_\theta(\hat{\theta}) = \frac{(n-2)(1-\theta^2)^{(n-1)/2}(1-\hat{\theta}^2)^{(n-4)/2}}{\pi} \int_0^\infty \frac{dw}{(\cosh w - \theta\hat{\theta})^{n-1}} \tag{4}$$

Uncertainty of Correlation Coefficient – How about the Prior?

Example

Option 1: Flat prior, $p(\theta) = 0.5$ on $\theta \in [-1, 1]$ and zero otherwise.
Option 2: Jeffreys' prior, $p(\theta) = \frac{1}{1-\theta^2}$.
Option 3: Triangular prior, $p(\theta) = 1 - |\theta|$.

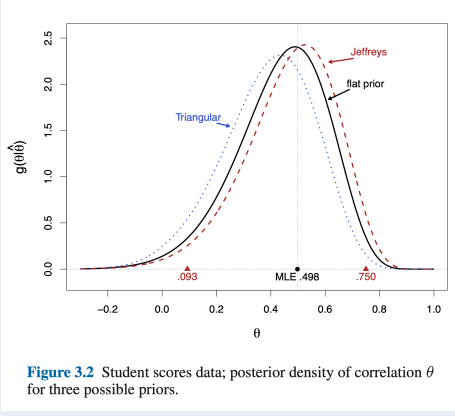


Figure 3.2 Student scores data; posterior density of correlation θ for three possible priors.

Uninformative Prior Distriutions

- Given a convincing prior distribution, Bayesian methods are more satisfactory than frequentist methods.
- A most influental method in the construction of priors is the employment **uninformative priors**.
- Laplace's principle of insufficient reason: **unform priors**
- Fisher 1920: Laplace's principle cannot be consistently used

Example

Consider a parameter change $\gamma = e^{\theta}$. Posterior probabilities such as $\Pr(\theta > 0|x)$ and $\Pr(\gamma > 1|x)$ would yield a different result assuming a uniform prior on γ or alternatively on θ. Neither choice is then uninformative!

Jeffreys' prior

For a **one-parameter family** $f_\mu(x)$, where the parameter space Ω is an interval of the real line $\mathbb{R}$, the Fisher information is

$$\mathcal{I}_\mu = \mathbb{E}\left[\left(\frac{\partial}{\partial \mu}\log f_\mu(x)\right)^2\right] \tag{5}$$

The **Jeffreys' prior** is given by

$$p(\mu) = \mathcal{I}_\mu^{1/2} \tag{6}$$

The $1/I_\mu$ is approximately the variance σ_μ^2 (why?), hence,

$$p(\mu) = \frac{1}{\sigma_\mu}. \tag{7}$$

Exercise

Calculate the Jeffreys' prior for the Poisson distribution.

Problems of the Frequentist Approach: Story of Metric reader

Example

Voltage mesurement on a batch of 12 tubees using a calibrated voltmeter

$$x \sim N(\mu, 1) \tag{8}$$

where x is any measurement and μ is the true batch voltage. The measurements range from 82 to 99, with a sample mean at $\bar{x} = 92$, which is reported as an unbiassed estimate.
Later it was discovered that any voltage exceeding 100 would have reported as $x = 100$. Due to frequentist point of view the result is no longer unbiassed but the estimate is “a little too small”—even though 100 was not in the sample set!

Problems of the Frequentist Approach: Story of Hypothesis testing

Example

Given an experiment where each month m one experiences

$$x_m \sim N(\mu, 1). \tag{9}$$

Goal: Test the null hypothesis $H_0 : \mu = 0$ against the right sided alternative $\mu > 0$.

Test statistic: $Z_m = \sum_{m=1}^{M} \frac{x_m}{\sqrt{M}} \sim N(0, 1)$.

At month 30, the scheduled end of the experiment, the p value is just under 0.05 risk level. Success!—Or is it?

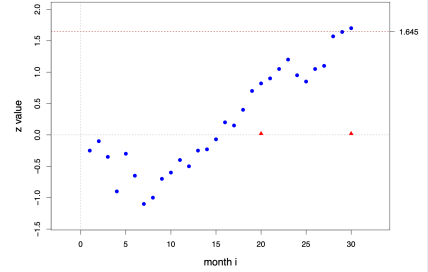


Figure 3.3 Z-values against null hypothesis $\mu = 0$ for months 1 through 30.

Problems of the Frequentist Approach: Selection bias

Example

Take a prostate cancer study consisting of 52 cases and 50 controls. Genetic activity measured for K=6033 genes.

Test statistic: for each gene the statistic x was computed s.t.

$$X_k \sim N(\mu_k), \quad k = 1, 2, \ldots, K \tag{10}$$

where μ_i is the true size of the gene.

Question: Interest on the highest values, what estimate give for μ_i?

Problem: Focusing on the largest of the 6033 produces an upward bias.

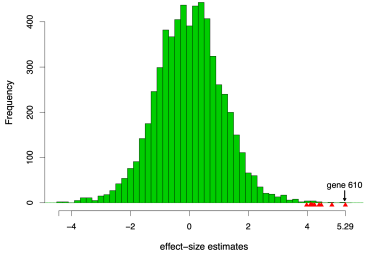


Figure 3.4 Unbiased effect-size estimates for 6033 genes, prostate cancer study. The estimate for gene 610 is $x_{610} = 5.29$. What is its effect size?

Frequentist vs. Bayesian Inference

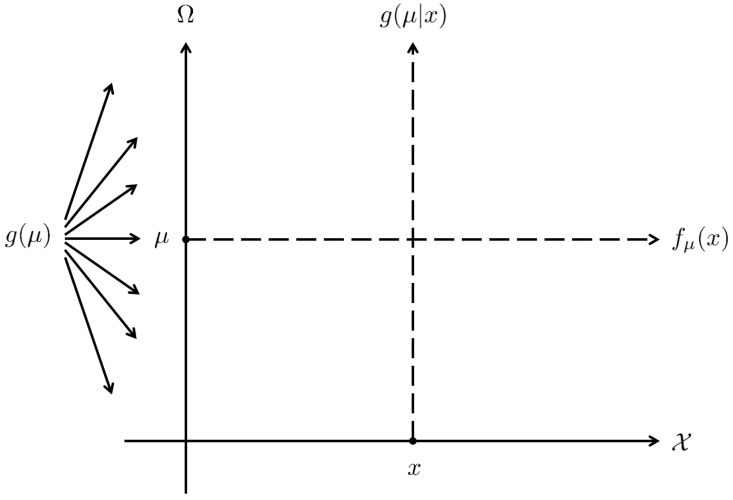


Figure 3.5 Bayesian inference proceeds vertically, given x; frequentist inference proceeds horizontally, given μ.

Frequentist vs. Bayesian Inference

Frequentist Approach

- Requires a choice of method suitable to the task
- The solution must be balanced between different family of distributions
- 'Hoping to do well' no matter what θ is.
- Different estimators for different questions.
- Claims to be the high ground for scientific objectivity,

Bayesian Approach

- Requires a prior distribution $p(\theta)$
- The posterior distribution is the complete solution for the problem
- Is 'all in' with the prior choice
- 'Answers all questions at once'
- In the absence of genuine prior information, results more subjective

Empirical Bayes: Robbins Formula

- An example where Bayesian formula for a single subject is estimated empirically i.e. frequentistcally.

Example

Table 6.1 *Counts y_x of number of claims x made in a single year by 9461 automobile insurance policy holders. Robbins' formula* (6.7) *estimates the number of claims expected in a succeeding year, for instance* 0.168 *for a customer in the $x = 0$ category. Parametric maximum likelihood analysis based on a gamma prior gives less noisy estimates.*

Claims x	0	1	2	3	4	5	6	7
Counts y_x	7840	1317	239	42	14	4	4	1
Formula (6.7)	.168	.363	.527	1.33	1.43	6.00	1.75	
Gamma MLE	.164	.398	.633	.87	1.10	1.34	1.57	

Problem: The incurance company wants to know for how many claims each policy holder will do in the *next* year

Robbins' Formula

- Let X_k is the number of claims made by policy holder in a year
- Assume that $X_k \sim \text{Poisson}(\theta_k)$, so θ_k is the expected value of the claims
- Assume an unknown prior $p(\theta)$.
- The expected value over the posterior yields

$$E\{\theta|x\} = \frac{\int_0^\infty [e^{-\theta}\theta^{x+1}/x!]p(\theta)d\theta}{\int_0^\infty [e^{-\theta}\theta^{x}/x!]p(\theta)d\theta} = \frac{(x+1)\int_0^\infty [e^{-\theta}\theta^{x+1}/(x+1)!]p(\theta)d\theta}{\int_0^\infty [e^{-\theta}\theta^{x}/x!]p(\theta)d\theta} \quad (11)$$

- On other hand the evidence

$$p(x) = \int_0^\infty p(x|\theta)p(\theta)d\theta = \int_0^\infty [e^{-\theta}\theta^{x}/x!]p(\theta)d\theta \quad (12)$$

thus,

$$E\{\theta|x\} = (x+1)\frac{p(x+1)}{p(x)} \quad (13)$$

which is known as **Robbins' formula**. Note: no prior knowledge required!

Empirical version of Robbins' formula

- The estimate for the evidence obtaind by the proportion of total counts in category x, or

$$p(\hat{x}) = \frac{y_x}{N}, \quad N = \sum_x y_x, \text{the total count}, \tag{14}$$

- So, $\hat{p}(0) = 7840/9461$, $\hat{p}(1) = 1317/9461$, etc.

- Hence.

$$\hat{E}\{\theta|\ x\} = (x+1)\frac{\hat{p}(x+1)}{\hat{p}(x)} = (x+1)\frac{y_{x+1}}{y_x}. \tag{15}$$

Robbin's formula vs. Gamma model

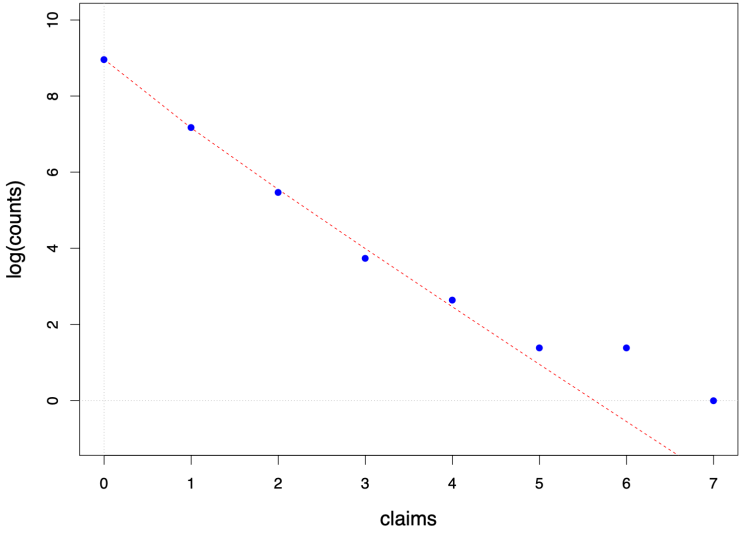


Figure 6.1 Auto accident data; log(counts) vs claims for 9461 auto insurance policies. The dashed line is a gamma MLE fit.

Empirical Bayes: Missing species problem

Example

Table 6.2 *Butterfly data; number y of species seen x times each in two years of trapping; 118 species trapped just once, 74 trapped twice each, etc.*

x	1	2	3	4	5	6	7	8	9	10	11	12
y	118	74	44	24	29	22	20	19	20	15	12	14
x	13	14	15	16	17	18	19	20	21	22	23	24
y	6	12	6	9	9	6	10	10	11	5	3	3

Corbet's question: If trapped for one additional year, how many new species would he expect to capture?

Empirical Bayes: Missing species problem (2)

- Suppose there are K species in all. The random variable X_k, the number of species k trapped in one time unit, we assume

$$X_k \sim \text{Poisson}(\theta_k), \quad k = 1, 2, \ldots, S. \tag{16}$$

- The number of spieces seen exactly x times is

$$Y_k = \#\{X_k = x\} \quad x = 1, 2, \ldots, 24. \tag{17}$$

- Now consider the further trapping period of $t = 1/2$ time units. We assume the number of spieces trapped in the new period

$$X_k(t) \sim \text{Poisson}(\theta_k t). \tag{18}$$

Empirical Bayes: Missing species problem (3)

- The probability that species k is *not* seen in the initial trapping period ($x_k = 0$) but is seen in the new period ($x_k(t) > 0$), is

$$P(X_k = 0 \wedge X_k(t) > 0) = e^{-\theta_k} \left(1 - e^{-\theta_k t}\right), \tag{19}$$

so that the expected number of new species in the new trapping period yields

$$E(t) = \sum_{k=1}^{K} e^{-\theta_k} \left(1 - e^{-\theta_k t}\right) \tag{20}$$

$$= K \int_0^{\infty} e^{-\theta} \left(1 - e^{\theta}\right) g(\theta) d\theta \tag{21}$$

$$= K \int_0^{\infty} e^{-\theta} \left[\theta t - \frac{(\theta t)^2}{2!} + \frac{(\theta t)^3}{3!} + \ldots\right] g(\theta) d\theta \tag{22}$$

where $g(\theta)$ is an empirical density putting probability 1/K on each θ_k values.

Empirical Bayes: Missing species problem (4)

- On the other hand, the expected value of Y_x

$$e_x = E(Y_x) = \sum_{k=1}^{K} \frac{e^{-\theta_k}\theta_k^x}{x!} \tag{23}$$

$$= K \int_0^\infty \left[\frac{e^{-\theta}\theta^x}{x!} \right] g(\theta) d\theta \tag{24}$$

$$\tag{25}$$

- Combining the two equations yields the surprising result

$$E(t) = e_1 t - e_2 t^2 + e_3 t^3 - \cdots \tag{26}$$

- We can estimate e_x:s by replacing them with the empirical estimates y_x that yields

$$\hat{E}(t) = y_1 t - y_2 t^2 + y_3 t^3 - \cdots \tag{27}$$

- As an answer to the Corbet's question $\hat{E}(1/2) = 45.2 \pm 11.2$.

Empirical Bayes: Missing species problem (5)

- The emprical assumption becomes unstable for $t > 1$—circumvent that Fisher used Gamma prior for θ

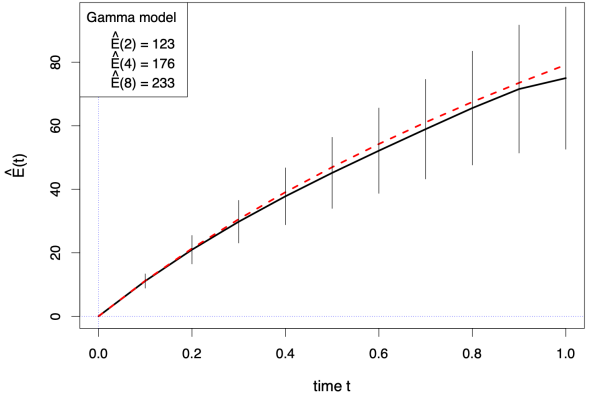


Figure 6.2 Butterfly data; expected number of new species in t units of additional trapping time. Nonparametric fit (solid) $\pm$ 1 standard deviation; gamma model (dashed).

Empirical Bayes: Medical Example

Example

- Cancer surgery can involve removal of surrounding lymph nodes in addition to the primary target
- Assumption: each patient k has some true probability θ_k of a node being positive. Assuming independence yields the **binomial model**

$$X_k \sim \text{Bin}(n_k, \theta_k). \tag{28}$$

  where X_k is the random variable desribing the number of positive nodes, n_k is the number of nodes removed.

- The ratios $P_k = X_k/n_k$, has the mean at $\text{E}\{P_k\} = \theta_K$ and variance $\text{Var}\{P_k\} = \theta_k(1 - \theta_K)/n_k$.

Empirical Bayes: Medical Example

Example

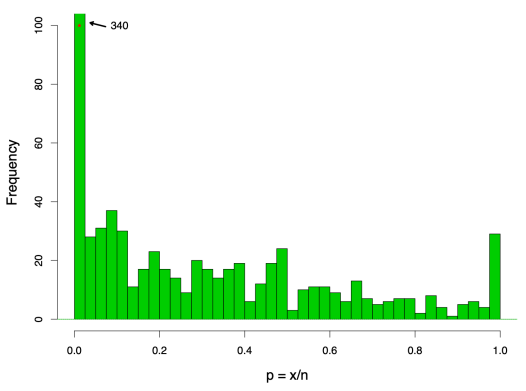


Figure 6.3 Nodes study; ratio $p = x/n$ for 844 patients; $n =$ number of nodes removed, $x =$ number positive.

Q: How should we estimate θ_k, using the data available in the Nodes study?

Empirical Bayes: Medical Example (2)

Solution:

- A Bayesian approach starts with a design of the prior $p(\theta)$.
- Assume a fourth order polynomial to $\log p(\theta)$, the bias (0th order term) from the constraint the $p(\theta)$ integrates to unity
- Maximise, with respect to the polynomial coefficients, the log marginal probability (likelihood function constructed from the marginals, ignorance on θ) sum over the patients.
- Construct the posterior distribution of θ_k

Empirical Bayes: Medical Example (3)

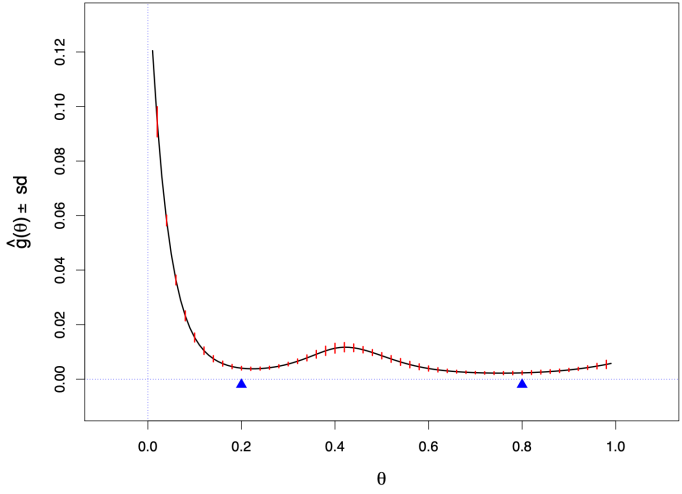


Figure 6.4 Estimated prior density $g(\theta)$ for the nodes study; 59% of patients have $\theta \leq 0.2$, 7% have $\theta \geq 0.8$.

Empirical Bayes: Medical Example (4)

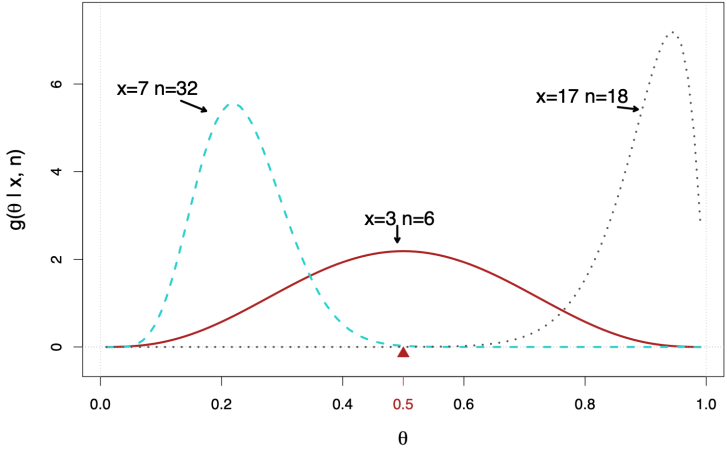


Figure 6.5 Empirical Bayes posterior densities of θ for three patients, given x = number of positive nodes, n = number of nodes.

Indirect Evidence

- Classic, clinical trials collect **direct** statistical evidence
- Bayesian inference provides theoretical basis to use **indirect** statistical evidence
- Emprical Bayes removes the Bayesian scaffolding, and allows to take the statistical evidence from large populations in reassuring priors

One More Example

A Bayesian Reconstruction Method with Marginalized Uncertainty Model for Camera Motion in Microrotation Imaging

Danai Laksameethanasan* and Sami S. Brandt

***Abstract*—Reconstruction of a 3-D structure from multiple projection images requires prior knowledge of projection directions or camera motion parameters that describe the relative positions and orientations of 3-D structure with respect to the camera. These parameters can be estimated using, for instance, the conventional correlation alignment and feature-based methods. However, the alignment methods are not perfect, where the inaccuracy of the estimated motion parameters causes artifacts in the reconstruction. To overcome this problem, we propose a Bayesian approach to reconstruct the object that takes the motion uncertainty distribution into account. Moreover, we consider the motion parameters as nuisance parameters and integrate them out from the posterior distribution, assuming a Gaussian uncertainty model, which yields a statistical cost function to be minimized. The proposed method is applied in microrotation fluorescence imaging, where we aim at 3-D reconstruction of a rotating object from an image series, acquired by an optical microscope. The experiments with simulated**

cage for cell trapping and producing rotational force for the cell rotation [1]–[3]. More precisely, the dielectric fields, generated by the electrodes, trap nonadherent cells in suspension inside the 3-D electrode cage and continuously rotate them the full 360° approximately around a chosen axis parallel to the fixed focal plane, as illustrated in Fig. 1. During the continuous rotation, the individual cells are sequentially imaged using an optical microscope, usually a fluorescence microscope. The advantage of the microrotation technique is that it offers a unique possibility to visualize nonadherent living cells from any orientations in suspension that provides a way for improving the 3-D optical resolution [4]–[6].To date, a major challenge in microrotation imaging is to provide fast, qualitative methods for reconstructing the rotating cells from a microrotation image series.

Quiz